

Cisco IOS Management & Upgrade Laboratory Report

Router Image Retrieval via TFTP & FTP Protocols with Boot Order Reconfiguration

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Target Devices: Cisco ISR 2911 (R1, R2) **Environment:** Cisco Packet Tracer

1. EXECUTIVE SUMMARY & TOPOLOGY OVERVIEW

This technical report documents the complete procedures and execution results for upgrading Cisco IOS software images across two ISR 2911 routers (**R1** and **R2**) from image version 15.1(4)M4 to 15.5(3)M4a. Network file transfers were executed using two primary protocols: **TFTP** (Trivial File Transfer Protocol) for R1, and **FTP** (File Transfer Protocol) for R2.

Lab Requirements & Objectives

- Task 1 (R1 via TFTP):** Transfer Cisco IOS image c2900-universalk9-mz.SPA.155-3.M4a.bin from Server **SRV1** (10.0.0.1) to R1's Flash storage using TFTP. Set boot system variable, verify execution, reload, and purge legacy image c2900-universalk9-mz.SPA.151-4.M4.bin.
- Task 2 (SRV1 FTP Config):** Verify / configure FTP user credentials on SRV1 (User: momo / Password: ccna) with Full Permissions (RWDNL).
- Task 3 (R2 via FTP):** Configure R2 with global FTP credentials, retrieve c2900-universalk9-mz.SPA.155-3.M4a.bin from SRV1 (10.0.0.1) via FTP, set boot path, reboot, and delete old image file from Flash.

2. ADDRESSING & DEVICE MATRIX

DEVICE	INTERFACE	IP ADDRESS	SUBNET MASK / PREFIX	ROLE / SERVICE
SRV1	FastEthernet0	10.0.0.1	255.255.255.0 (/24)	TFTP Server & FTP Server (User: momo / Pass: ccna)
SW1	VLAN 1 / Ports	Unmanaged / Trunk	10.0.0.0/24	Layer 2 Access Switch
R1	GigabitEthernet0/1	10.0.0.254	255.255.255.0 (/24)	LAN Gateway / TFTP Client Target
R1	GigabitEthernet0/0	192.168.12.1	255.255.255.252 (/30)	P2P WAN Link to R2
R2	GigabitEthernet0/0	192.168.12.2	255.255.255.252 (/30)	Remote Router / FTP Client Target

3. EXECUTION: R1 IOS UPGRADE VIA TFTP

Interface GigabitEthernet0/1 was brought up on R1 to establish connectivity with TFTP Server 10.0.0.1. The IOS image was transferred and set as the boot system image.

R1 - Interface Activation & TFTP Transfer Command Sequence

```
R1(config-if)# no shut
R1(config-if)# ex
R1# copy tftp: flash:
Address or name of remote host []? 10.0.0.1
Source filename []? c2900-universalk9-mz.SPA.155-3.M4a.bin
Destination filename [c2900-universalk9-mz.SPA.155-3.M4a.bin]?

Accessing tftp://10.0.0.1/c2900-universalk9-mz.SPA.155-3.M4a.bin...
Loading c2900-universalk9-mz.SPA.155-3.M4a.bin from 10.0.0.1:
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK - 33591768 bytes]

33591768 bytes copied in 5.547 secs (635838 bytes/sec)
```

R1 - Boot System Configuration & Flash Directory Verification

```
R1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# boot system flash:c2900-universalk9-mz.SPA.155-3.M4a.bin
R1(config)# ex
R1# wr
Building configuration...
[OK]
R1# show flash

System flash directory:
File Length Name/status
  3 33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
  4 33591768 c2900-universalk9-mz.SPA.155-3.M4a.bin
  2 28282 sigdef-category.xml
  1 227537 sigdef-default.xml
[67439355 bytes used, 188304645 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

R1 - Post-Reload Verification (show version)

```
R1> en
R1# show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.5(3)M4a, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Compiled Thu 06-Oct-16 14:43 by mnguyen

ROM: System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
cisco1941 uptime is 27 seconds
System returned to ROM by power-on
System image file is "flash0:c2900-universalk9-mz.SPA.155-3.M4a.bin"
Last reload type: Normal Reload
```

4. R1 LEGACY IOS IMAGE PURGE & TROUBLESHOOTING ANALYSIS

Troubleshooting Insight (Observed Syntax Errors during Lab):

Initial attempts to delete the old image failed due to copying the byte size parameter into the filename string (delete flash: 33591768c2900...) and passing invalid confirmation text (yes) directly to the prompt. Correct execution requires specifying only the clean filename and pressing **Enter** to confirm default prompts.

R1 - Command Errors & Successful Legacy File Purge

```
R1# delete flash:33591768c2900-universalk9-mz.SPA.151-4.M4.bin
Delete filename [33591768c2900-universalk9-mz.SPA.151-4.M4.bin]?
Delete flash:/33591768c2900-universalk9-mz.SPA.151-4.M4.bin? [confirm]
%Error deleting flash:/33591768c2900-universalk9-mz.SPA.151-4.M4.bin (No such file or directory)

R1# delete flash:c2900-universalk9-mz.SPA.151-4.M4.bin
Delete filename [c2900-universalk9-mz.SPA.151-4.M4.bin]? yes
Delete flash:/yes? [confirm]
%Error deleting flash:/yes (No such file or directory)

R1# delete flash:c2900-universalk9-mz.SPA.151-4.M4.bin
Delete filename [c2900-universalk9-mz.SPA.151-4.M4.bin]? [Press Enter]
Delete flash:/c2900-universalk9-mz.SPA.151-4.M4.bin? [confirm] [Press Enter]

R1# show flash
System flash directory:
File Length Name/status
  4  33591768 c2900-universalk9-mz.SPA.155-3.M4a.bin
  2   28282 sigdef-category.xml
  1  227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total] <--- Free Flash space increased from
188MB to 221MB
```

5. EXECUTION: R2 IOS UPGRADE VIA FTP

Unlike TFTP (which is connectionless UDP and unauthenticated), FTP requires authenticating credentials before file transmission. Global IP FTP parameters were defined on **R2** to match SRV1's user configuration.

SRV1 FTP Service Configuration Matrix:

- **Username:** momo
- **Password:** ccna
- **Permissions Enabled:** Read (R), Write (W), Delete (D), Name (N), List (L)

R2 - FTP Global Credential Setup & File Transfer

```
R2> en
R2# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)# ip ftp username momo
R2(config)# ip ftp password ccna
R2(config)# ex

R2# copy ftp: flash:
Address or name of remote host []? 10.0.0.1
Source filename []? c2900-universalk9-mz.SPA.155-3.M4a.bin
Destination filename [c2900-universalk9-mz.SPA.155-3.M4a.bin]?

Accessing ftp://10.0.0.1/c2900-universalk9-mz.SPA.155-3.M4a.bin...
[OK - 33591768 bytes]

33591768 bytes copied in 102.819 secs (34302 bytes/sec)
```

R2 - Boot System Reconfiguration, Directory Verification & Reload

```
R2# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)# boot system flash:c2900-universalk9-mz.SPA.155-3.M4a.bin
R2(config)# do sh flash

System flash directory:
File Length Name/status
  3 33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
  4 33591768 c2900-universalk9-mz.SPA.155-3.M4a.bin
  2 28282 sigdef-category.xml
  1 227537 sigdef-default.xml
[67439355 bytes used, 188304645 available, 255744000 total]

R2(config)# do wr
Building configuration...
[OK]
R2(config)# ex
R2# reload
Proceed with reload? [confirm]
```

R2 - Post-Reload Verification & Legacy Image Cleanup

```
R2> en
R2# delete flash:c2900-universalk9-mz.SPA.151-4.M4.bin
Delete filename [c2900-universalk9-mz.SPA.151-4.M4.bin]? [Press Enter]
Delete flash:/c2900-universalk9-mz.SPA.151-4.M4.bin? [confirm] [Press Enter]

R2# sho flash

System flash directory:
File Length Name/status
  4 33591768 c2900-universalk9-mz.SPA.155-3.M4a.bin
  2 28282 sigdef-category.xml
  1 227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

6. TECHNICAL COMPARISON & CONCLUSION

PROTOCOL CHARACTERISTIC	TFTP TRANSFER (R1)	FTP TRANSFER (R2)
Transport Layer Protocol	UDP (Port 69)	TCP (Ports 20 & 21)
Authentication	None (Anonymous)	User Credentials (`momo` / `ccna`)
Transfer Speed (in Lab)	~635,838 bytes/sec (5.547 seconds)	~34,302 bytes/sec (102.819 seconds)
Reliability Mechanism	Lock-step Application ACKs	TCP Windowing & Error Recovery
Final OS Version Status	VERSION 15.5(3)M4A ACTIVE	VERSION 15.5(3)M4A ACTIVE

Conclusion & Results:

Both Cisco 2911 routers were successfully upgraded to IOS version 15.5(3)M4a. The legacy IOS files were purged from Flash storage on both units, successfully restoring over 33.5 MB of Flash memory space on each device. The boot path settings were verified persistent across device reboots.